

Residential House Price Prediction - Data Preprocess

Dataset Inspection:

The dataset was loaded and examined to understand its structure, dimensions, data types, and feature information. The dataset contains 1,460 records and 81 features related to residential properties and their sale prices.

Missing Value Analysis:

A detailed missing value analysis was performed to identify features containing null values. Several numerical and categorical features were found to have missing observations.

Missing Value Treatment:

The **LotFrontage** feature contained 259 missing values. Since the distribution exhibited outliers and was positively skewed, missing values were imputed using the **median** rather than the mean. This approach minimizes the impact of extreme values and preserves the distribution of the feature.

For categorical features such as **BsmtQual**, **BsmtCond**, **GarageType**, **FireplaceQu**, **PoolQC**, and **Fence**, missing values represented the absence of the corresponding facility rather than missing information. Therefore, these values were replaced with "**None**" to accurately reflect the property's characteristics.

For categorical features where missing values were due to incomplete records, such as **Electrical** and **MasVnrType**, the **mode** was used for imputation.

Duplicate Record Check:

A duplicate record analysis was performed on the dataset. No duplicate entries were identified; therefore, no records were removed.

Outlier Detection and Treatment:

Outliers in the numerical features were identified using the Interquartile Range (IQR) method. Extreme values outside the lower and upper limits were treated using outlier replacement (capping), where values were replaced with the calculated boundary limits instead of removing records. This approach preserves dataset size while minimizing the influence of extreme observations.

Data Validation:

After preprocessing, the dataset was re-evaluated to confirm that:

- Missing values were successfully handled.
- Duplicate records were removed.
- Outliers were treated appropriately.
- Data quality was suitable for further analysis.

Exporting the Cleaned Dataset:

The pre-processed dataset was saved as a new CSV file (house_price_cleaned.csv) for use in exploratory data analysis, feature engineering, and model development.